

COURSE SYLLABUS محتويات المقرر

الفصل الدراسي الأول

P111 & PP1

Properties of Matters

2 h lect./w + 1 h lab ./w + 1/2 h lut /w

1st year, 1st semester (for Science & Math. students)

وشعبة التعليم الابتدائي

ف ١١١، ف أ ١

خواص مادة

COURSE NO.

COURSE TITLE

NUMBER OF HOURS

YEAR & SEMISTER

COURSE DESCRIPTION

COURSE GOALS:

PREREQUESITES AND COREQUISITES : متطلبات السابقة أو المتوازية :

١. رقم المقرر :

٢. عنوان المقرر :

٣. عدد الساعات :

٤. الفرقة والفصل الدراسي :

٥. وصف المقرر :

٦. أهداف المقرر

1- Students have to know dimensional analysis and its applications.

2- Students have to know physical phenomena related to properties of matter and its applications in daily life.

The course introduces students the basic physical quantities, standard units and dimensional analysis. Also it introduces some physical concept in properties of matter and their uses in daily life.

التقويم Assessment	المصادر التعليمية Educational Resources	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	الزمن المتوقع لكل موضوع	الموضوع Topic	مخرجات التعلم Intended Learning Outcomes	م
- Semester activities including classroom interactions and Quizzes. - Mid-term exam - Lab performance evaluation. - Oral exam. - Final exam.	- Computer simulation programs and slides. - Transparences. - Manual of solved problems (answer and solutions) - Text books: 1- Fundamentals of physics by D-Halliday & R.Resnik. 2- Physics for scientists and Engineers by R. Serway & R. Beichner. 3- Physics by J.W. Kane & M.M. Sternhein	- Lectures. - Laboratory experiments. - Problems and essay assignments.	6h	1- Dimensional analysis and application.	The student have to recognize the following: 1- Basic physical quantities and dimensional analysis with emphasis on validation of basic physical relations and how to deduce basic physical relation. 2- Elasticity of materials, surface tension, capillarity, basic connected relations and applications. 3- Fluids dynamics, viscosity of liquids, diffusion and osmotic phenomenon	
			4h	2- Elasticity and elastic modules.		
			4h	3- Surface tension and capillarity.		
			4h	4- Hydrodynamics.		
			4h	5- Bernoulli's equation & appl.		
			2h	6- Viscosity.		
			2h	7- Diffusion and osmotic phenomenon		

التقويم Assessment	المصادر التعليمية Educational Resources	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	الزمن المتوقع لكل موضوع	الموضوع Topic	مخرجات التعلم Intended Learning Outcomes	م
- Semester activities including classroom interactions and Quizzes. - Mid-term exam - Lab performance evaluation. - Oral exam. - Final exam.	- Computer simulation programs and slides. - Transparencies. - Manual of solved problems (answer and solutions) - Text books: 1- Fundamentals of physics by D-Halliday & R.Resnik. 2- Physics for scientists and Engineers by R. Serway & R. Beichner. 3- Physics by J.W. Kane & M.M. Sternhein	- Lectures. - Laboratory experiments. - Problems and essay assignments.	6h	1- Dimensional analysis and application.	The student have to recognize the following: 1- Basic physical quantities and dimensional analysis with emphasis on validation of basic physical relations and how to deduce basic physical relation. 2- Elasticity of materials, surface tension, capillarity, basic connected relations and applications. 3- Fluids dynamics, viscosity of liquids, diffusion and osmotic phenomenon	
			4h	2- Elasticity and elastic modules.		
			4h	3- Surface tension and capillarity.		
			4h	4- Hydrodynamics.		
			4h	5- Bernoulli's equation & appl.		
			2h	6- Viscosity.		
			2h	7- Diffusion and osmotic phenomenon		

COURSE SYLLABUS محتويات المقرر

الفصل الدراسي الأول

©

P112 & PP2

Electricity and Magnetism

2 h lect./w + 1h lab./w + 1/2 h tut /w

1st year, 1st semester (For Science and Math. Sections)

وسّعة التعليم الابتدائي

ف ١١٢، ف أ ٢

كهربية ومغناطيسية

COURSE NO.

COURSE TITLE

NUMBER OF HOURS

YEAR & SEMISTER

COURSE DESCRIPTION

COURSE GOALS:

PREREQUISITES AND COREQUISITES : متطلبات السابقة أو المتوازية :

١. رقم المقرر :

٢. عنوان المقرر :

٣. عدد الساعات :

٤. الفرقة والفصل الدراسي :-

٥. وصف المقرر :

٦. أهداف المقرر

- 1- Students have to understand the basic concepts of electricity and its magnetic effects.
- 2- The ability to apply electricity and magnetism concepts in different sciences

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م	مخرجات التعلم Intended Learning Outcomes	الموضوع Topic	الزمن المتوقع لكل موضوع	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	المصادر التعليمية Educational Resources	التقويم Assessment
	The student have to know the following: 1- Calculation of the forces between charges. 2- Electric fields, potential together with magnetic fields and their application in everyday life. 3- Basic components of electric circuits and their role in achieving the required electric circuit out put 4- Electric and magnetic properties of materials.	1- Coulomb's law. 2- The electric field. 3- Current, resistance, resistivity. 4- Line of forces. 5- The electric potential. 6- Circuits. 7- Properties of dielectrics. 8- Capacitance and Capacitors. 9- Magnetic field strength. 10- Magnetic field of a current and of a moving charge.	2hr 2hr 2hr 2hr 4hr 2hr 4hr 4hr 2hr 2hr	- Lectures. - Laboratory experiments. - Problems and essay assignments.	- Computer simulation programs and slides. - Transparencies. - Manual of solved problems (answer and solutions) - Text books: 1- "Fundamentals of Physics by David Halliday and Robert Resnick. John Wiley & Sons. 2- Physics by Raymond A. Serway, Robert J. Beichner, John W. Jewett. Saunders College publishing"	- Semester activities including classroom interactions and Quizzes. - Mid-term exam - Lab performance evaluation. - Oral exam. - Final exam.

COURSE SYLLABUS محتويات المقرر

الفصل الدراسي الثاني

(٨)

P121, PP3

ف ١٢١، ف ٣ أ

COURSE NO.

١. رقم المقرر :

Heat

حرارة

COURSE TITLE

٢. عنوان المقرر :

2 h lect./w + 1 h lab./w + 1/2 h tut /w

NUMBER OF HOURS

٣. عدد الساعات :

1st year, 2nd semester (For Science & Math. Sections)

YEAR & SEMISTER

٤. الفرقة والفصل الدراسي:-

وشعبة التعليم الابتدائي

COURSE DESCRIPTION

٥. وصف المقرر :

The course introduces the students to the phenomena associated with heat and its effect on materials and phase transformations. This course also studies the mechanism of heat transfer and microscopic description.

COURSE GOALS:

٦. أهداف المقرر

Study of heat phenomena and its effect on materials and fundamental physical quantities in thermal systems

٧. متطلبات السابقة أو المتوازية : PREREQUESITES AND COREQUISITES

التقويم Assessment	المصادر التعليمية Educational Resources	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	الزمن المتوقع لكل موضوع	الموضوع Topic	مخرجات التعلم Intended Learning Outcomes	م
- Semester activities including classroom interactions and Quizzes. - Mid-term exam - Lab performance evaluation. - Oral exam. - Final exam.	- Computer simulation programs and slides. - Transparencies. - Manual of solved problems (answer and solutions) - Text books: 1- " Physics for Scientists and Engineering ", F.W Sears and R.J Beichner, Saunders college publishing. 2- " Physics Principles with applications ", D.C Giancoli, USA. 3- " University physics", F.W. Sears, M.W. Zemasky and H.D. Young, Wesley series in physics,	- Lectures. - Laboratory experiments. - Problems and essay assignments.	4hr 2hr 6hr 6hr 2hr 2hr	1- Concept of heat, temperature and thermal equilibrium. 2- Mechanical equivalent of heat. 3- Effect of heat on materials: Change of state (phase rule and phase diagram), change in size (thermal expansion with some applications and anomalous behavior of water), thermal stresses and change in color. 4- Mechanisms of heat transfer (simple and compound walls) and thermal conductivity. 5- Kinetic theory of gases. 6- Kinetic calculation of pressure and	The student have to recognize the following: 1- Concept of heat as a form of energy together with concept of temperature and its measurements. 2- Effect of heat on different materials and the mechanisms of heat transfer through them. 3- Basic postulate of kinetic theory of gases are successful in	

التقويم Assessment	المصادر التعليمية Educational Resources	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	الزمن المتوقع لكل موضوع	الموضوع Topic	مخرجات التعلم Intended Learning Outcomes	م
	USA		2hr 2hr	temperature. 7- Maxwell-Boltzmann distribution of molecular speed. 8- Thermal analysis (DTA and DSC)	microscope description of gas properties.	

COURSE SYLLABUS المحتويات المقرر

الفصل الدراسي الثاني

P122& PP4	ف ١٢٢ & ف أ ٤	<u>COURSE NO.</u>	١. رقم المقرر :
Geometrical Optics	بصريات هندسية	<u>COURSE TITLE</u>	٢. عنوان المقرر :
2 h lect./w + 1h lab./w + 1/2 h tut /w		<u>NUMBER OF HOURS</u>	٣. عدد الساعات :
1 st year, 2 nd semester (Science & Math Sections)		<u>YEAR & SEMISTER</u>	٤. الفرقة والفصل الدراسي :-
وشعبة التعليم الابتدائي		<u>COURSE DESCRIPTION</u>	٥. وصف المقرر :

The course introduces the students to the nature and the theories of light – Also students have to recognize the fundamentals of light reflection and refraction at different surfaces and applications of them in connection with optical instruments

COURSE GOALS:

٦. أهداف المقرر

The Student should understand the theories of light and their applications in optical instruments

٧. متطلبات السابقة أو المتوازية : PREREQUESITES AND COREQUISITES

التقويم Assessment	المصادر التعليمية Educational Resources	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	الزمن المتوقع لكل موضوع	الموضوع Topic	مخرجات التعلم Intended Learning Outcomes	م
- Semester activities including class room interactions and Quizzes. - Mid-term exam - Lab performance evaluation. - Oral exam. - Final exam.	- Computer simulation programs and slides. - Transparencies. - Manual of solved problems (answer and solutions - Text books: 1- Fundamental of optics by F.A Jenkins & H.E White. - والترجمة العربية أساسيات الفيزياء، ترجمة أ.د عبد الفتاح احمد الشاذلي 2- University physics by F.W.Sears & M.W Zemansky. 3- Fundamental of physics by D.Halliday & R. Resnik.	- Lectures. - Laboratory experiments. - Problems and essay assignments.	6hr 6hr	1- The nature and propagation of light. - Nature of light - Sources of light. - Speed of light. - Theories of light. - Dopler effect. 2- Reflection and refraction at plane surfaces. - Derivation of the laws of reflection and refraction from corpuscular principle. - Derivation of the laws of reflection and refraction from Huygen's principle. - Snell's law. - Total reflection. - Refraction by a prism - Dispersion - Rain bow - Brewster's law	1- The student has to recognize nature of theories of light. 2- To be able to interpret reflection and refraction of light at plane surfaces from both points of view of corpuscular and wave theories.	

التقويم Assessment	المصادر التعليمية Educational Resources	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	الزمن المتوقع لكل موضوع	الموضوع Topic	مخرجات التعلم Intended Learning Outcomes	م
	4- Physics for scientists and Engineers by R.A Serway & R.J Beichner. 5- Optical physics by S.Lipson, H.Lipon & D.S Tannhause. 6- التداخل في الضوء والأطياف ، أ.د نايل بركات أ.د احمد امين حمزة		6hr 2hr 2hr 2hr 2hr	3-Reflection and refraction of spherical waves at surfaces. - Spherical wave - Plane mirror. - Spherical wave. - Spherical surface. (reflection). 4- Thin lenses. 5- Thick lenses. 6- Lens Aberrations. 7- Camera. 8- The eye. 9- Microscopes. 10- Telescopes.	3- To be able to interpret reflection and refraction of spherical waves on different surfaces. 4- To recognize operation of different kinds of lenses and their applications.	

COURSE SYLLABUS محتويات المقرر

الفصل الدراسي الأول

(2)

P 211	211 ف	<u>COURSE NO.</u>	1. رقم المقرر :
Thermodynamics	ديناميكا حرارية	<u>COURSE TITLE</u>	2. عنوان المقرر :
2 h lect./w + 2h lab./w		<u>NUMBER OF HOURS</u>	3. عدد الساعات :
2 nd year, 1 st semester (for Physics & Math. Sections)		<u>YEAR & SEMISTER</u>	4. الفرقة والفصل الدراسي :-
		<u>COURSE DESCRIPTION</u>	5. وصف المقرر :

The course introduces the students to the principles of thermo dynamical systems, processes, laws and applications.

COURSE GOALS:

6 . أهداف المقرر

The students will be trained on applications of conservation of energy within the context of thermodynamics

PREREQUISITES AND COREQUISITES : متطلبات السابقة أو المتوازية :

التقويم Assessment	المصادر التعليمية Educational Resources	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	الزمن المتوقع لكل موضوع	الموضوع Topic	مخرجات التعلم Intended Learning Outcomes	م
- Semester activities including class room interactions and Quizzes. - Mid-term exam - Lab performance evaluation. - Oral exam. - Final exam.	- Computer simulation programs and slides. - Transparencies. - Manual of solved problems (answer and solutions) - Text Books 1- "Heat & Thermodynamics" Zemansky & dittman, McGrew- Hill Company. 2- " Physics: Principles with applications", d.c Giancoli, USA. 3- " University Physics" F.W Sears, M.W Zemasky and H.D Young, Wesley Series in physics.	- Lectures. - Laboratory experiments. - Problems and essay assignments	1hr 1hr 2hr 2hr 2hr 2hr 2hr	1- Thermodynamic systems and static variables. 2- Types of thermodynamic processes. 3- Thermodynamic equilibrium. 4- Work in thermodynamic. 5- Internal energy of a gas. 6- Point and path functions. 7- First law of thermodynamics and conservation of energy. 8- Applications of the first law in different processes.	The students have to recognize the following: 1- Systems, state variables and thermodynamical processes. 2- Work, internal energy and first law of thermodynamics and its applications in different processes.	

م	مخرجات التعلم Intended Learning Outcomes	الموضوع Topic	الزمن المتوقع لكل موضوع	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	المصادر التعليمية Educational Resources	التقويم Assessment
	3- Concepts of enthalpy, surface energy and transformation of heat into work and the associated applications. 4- Second law of thermodynamics, concept of entropy and the associated processes.	9- Enthalpy. 10- Surface energy. 11- Transformation of heat into work and thermal efficiency. 12- Heat engines. 13-Carnot cycle and refrigerator. 14-Second law of thermodynamics. 15- Entropy in reversible and irreversible processes. 16-Thermodynamic function	1 hr 1 hr 1 hr 1 hr 2 hr 2 hr 2 hr 2 hr			

COURSE SYLLABUS المحتويات المقرر

الفصل الدراسي الأول

P212, PP5

Vibrations & Waves

2 h lect./w + 2h lab./w

2nd year, 1st Semester (for Physics & Math. Sections)

+ شعبة التعليم الابتدائي.

ف أ 5، ف 212 COURSE NO.

اهتزازات وموجات COURSE TITLE

NUMBER OF HOURS

YEAR & SEMISTER

COURSE DESCRIPTION

COURSE GOALS:

1. رقم المقرر :

2. عنوان المقرر :

3. عدد الساعات :

4. الفرقة والفصل الدراسي:

5. وصف المقرر :

6 . أهداف المقرر

<u>المخرجات المتوقعة</u> INTENDED LEARNING OUTCOMES	<u>الأهداف</u> GOALS
The students have to deduce: 1- The differential equation of vibrational motion. 2- The differential equation of wave motion. 3- The distribution of velocities and pressure in plane wave.	The students have to know the nature of vibrational wave motions with emphasis on their mathematical descriptions and experimental verifications.

7. متطلبات السابقة أو المتوازية : PREREQUISITES AND COREQUISITES

م	مخرجات التعلم Intended Learning Outcomes	الموضوع Topic	الزمن المتوقع لكل موضوع	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	المصادر التعليمية Educational Resources	التقويم Assessment
	The students have to recognize the following: 1- Simple harmonic motion and their applications 2- Superposition of more than one S.H.M and the associated Lissajou's figures.	1- Simple Harmonic Motion: Differential equation and graphical representation. 2- Energy of vibration. 3- Oscillation with one and two degrees of freedom. 4- Linearity and superposition principle. 5- Simple and compound pendulum. 6- Simple harmonic oscillation of a mass between springs and of loaded spring. 7- Lissajou's figures. 8- Composition of two simple harmonic oscillations.	2hr 1hr 2hr 2hr 2hr 2hr 1hr 2hr	- Lectures. - Problems assignments and essay	- Computer simulation programs and slides. - Transparencies. - Manual of solved problems (answer and solutions) - Text books: 1- Fundamentals of optics, F.A.Jenkins and H.E White. 2- " أساسيات الضوء " ترجمة أ.د عبد الفتاح احمد الشاذلي 3- " Light", R.W. Ditchburn	- Semester activities including class room interactions and Quizzes. - Mid-semester exam - Final exam. - Oral Exam. - Lab performance evaluation.

التقويم Assessment	المصادر التعليمية Educational Resources	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	الزمن المتوقع لكل موضوع	الموضوع Topic	مخرجات التعلم Intended Learning Outcomes	م
			2hr	9- Free, forced and resonant vibration.	3- Free, forced and resonant vibrations. (P.T.O).	
			2hr	10- Damped and un-damped vibration.		
			1hr	11- What does propagate in wave motion?	4- Wave motion characteristics, types of waves.	
			2hr	12- Characteristics of wave motion: Transverse and Longitudinal wave motion.		
			1hr	13- Differential equation of wave motion.		
			1hr	14- Particle and wave velocities.		
			2hr	15- Distribution of velocities and pressure wave.		
			1hr	16- Energy of Progressive wave.		

COURSE SYLLABUS محتويات المقرر

الفصل الدراسي الأول.

PP6 & P213

Astrophysics and Meteorology

2 h lect. Weekly

2nd year & 1st Semester (For Physics Section)

+ شعبة التعليم الابتدائي.

ف أ 6 ، ف 213

فيزياء فلكية

COURSE NO.

COURSE TITLE

NUMBER OF HOURS

YEAR & SEMISTER

COURSE DESCRIPTION

COURSE GOALS:

7. متطلبات السابقة أو المتوازية : PREREQUISITES AND COREQUISITES

1. رقم المقرر :

2. عنوان المقرر :

3. عدد الساعات :

4. الفرقة والفصل الدراسي:-

5. وصف المقرر :

6 . أهداف المقرر

The courses introduce the student to basis principles of astronomy and meteorology.

* Students have to know principles of astronomy, astrophysics and meteorology.

م	مخرجات التعلم Intended Learning Outcomes	الموضوع Topic	الزمن المتوقع لكل موضوع	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	المصادر التعليمية Educational Resources	التقويم Assessment
	Students have to recognize the following: 1- The role of observational astronomy. 2- Kinds of stars, stars generations and collapse. 3- The Sun and solar system. 4- Composition, structure and behavior of atmosphere.	1- Historical introduction, astronomical units and observational astronomy. 2- Pulsating and rotating stars- Energy transport and generation of stars- star formation- stellar evaluation. 3- The sun (structure – Stellar phenomena). 4- Atmosphere (composition, structure and behavior) synoptic weather analysis principles, techniques, structure and development of weather systems.	4hr 6hr 4hr 4hr	- Lectures. - Problems and essay assignments - Sight Seeing	- Computer simulation programs and slides. - Transparencies. - Manual of solved problems (answer and solutions) - Text books. 1- آفاق علم الفلك، تأليف أ.د محمد احمد سليمان. 2- علم الفلك العام، تأليف د. ميرفت السيد عوض، د. مصطفى كمال محمود	- Semester activities including class room interactions and Quizzes. - Mid-term exam. - Final exam. - Oral Exam.

م	مخرجات التعلم Intended Learning Outcomes	الموضوع Topic	الزمن المتوقع لكل موضوع	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	المصادر التعليمية Educational Resources	التقويم Assessment
	5- Weather analysis and fundamentals of modern meteorology.	5- Weather systems- Fundamentals of modern meteorology- interstellar dynamic structure.	4hr			
	6- Role of space ships and satellites (communication and meteorology)	6- Role of space- ships and satellites in communications and meteorology.	4hr			

COURSE SYLLABUS محتويات المقرر

الفصل الدراسي الثاني.

P214& PP8	للمبانيات والكيمياء للتعليم الابتدائي ف أ 8، ف 214	<u>COURSE NO.</u>	1. رقم المقرر :
Modern physics	فيزياء حديثة	<u>COURSE TITLE</u>	2. عنوان المقرر :
2 h lect./w + 1h (lab for math)(tut for ch.)		<u>NUMBER OF HOURS</u>	3. عدد الساعات :
الفصل الدراسي الأول	الفرقة الثانية (شعبة الرياضيات)	<u>YEAR & SEMISTER</u>	4. الفرقة والفصل الدراسي :-
الفصل الدراسي الثاني	الفرقة الثانية (شعبة الكيمياء)		
	شعبة التعليم الابتدائي - علوم		
	<u>COURSE DESCRIPTION</u>	5. وصف المقرر :	
The course introduces the students the revolution principles of physics at the beginning of the 20 th Century			
	<u>COURSE GOALS:</u>	6. أهداف المقرر	
* Developing a clear understanding of the basic concepts of physics needed to study all topics in modern physics and applied mathematics and quantum Chemistry.			
<u>7. متطلبات السابقة أو المتوازية : PREREQUESITES AND COREQUISITES</u>			

التقويم Assessment	المصادر التعليمية Educational Resources	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	الزمن المتوقع لكل موضوع	الموضوع Topic	مخرجات التعلم Intended Learning Outcomes	م
- Semester activities including class room interactions and Quizzes. - Mid-semester exam - Oral Exam. - Final exam.	- Computer simulation programs and slides. - Transparencies. - Manual of solved problems (answer and solutions) - <u>Text books</u> 1- Concepts of Modern Physics, By Arthur Beiser McGrew Hill. 2- Modern Physics by Eisberg	- Lectures. - Problems assignments and essay	8hr.	1- Particle nature of waves: Photo- electric effect- Photon theory of light- Wave particle duality- X-ray- X-ray diffraction- The Compton effect.	1- Particle nature of waves and the experiments confirming this nature.	
			6hr	2- Wave nature of particles: De-Broglie wave and velocity- Phase and group velocity- Electron diffraction.	2- Wave nature of particles and the experiments confirming this nature.	
			6hr	3- Heizenberg uncertainty principle- Uncertainty of (position- momentum) and (Energy- time)- Using uncertainty principle to prove the nonexistence of electron inside the nucleus.	3- Heizenberg uncertainty principle and its applications	

التقويم Assessment	المصادر التعليمية Educational Resources	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	الزمن المتوقع لكل موضوع	الموضوع Topic	مخرجات التعلم Intended Learning Outcomes	م
			6hr	4- Atomic Structure Rutherford scattering- nuclear dimensions and atomic dimensions- Electron orbits- atomic spectra- Bohr's atom- Atomic excitation	4- Atomic structure and spectra	

COURSE SYLLABUS محتويات المقرر

الفصل الدراسي الثاني

P 221

Alternating Current

2 h lect./w + 1h lab./w

2nd year, 2nd semester

COURSE NO.

تيار متردد COURSE TITLE

NUMBER OF HOURS

YEAR & SEMISTER

COURSE DESCRIPTION

١. رقم المقرر :

٢. عنوان المقرر :

٣. عدد الساعات :

٤. الفرقة والفصل الدراسي :

٥. وصف المقرر :

The course introduces the students to the principles of attending current circuits that includes the basic circuit elements and their applications

COURSE GOALS:

٦. أهداف المقرر

* The students have to know how A.C circuits are built up with emphases on filters, transformers and power transmission

٧. متطلبات السابقة أو المتوازية : PREREQUISITES AND COREQUISITES

التقويم Assessment	المصادر التعليمية Educational Resources	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	الزمن المتوقع لكل موضوع	الموضوع Topic	مخرجات التعلم Intended Learning Outcomes	م
- Semester activities including class room interactions and quizzes. - Mid-term exam - Lab performance evaluation. - Oral exam. - Final exam.	- Computer simulation programs and slides. - Transparencies. - Manual of solved problems (answer and solutions) - Text Books 1- Physics by Joseph W.Kane, Morton M. Sternhein. John wiley and sons. 2- Physics for scientists and engineers. By Raymond A.Serway, Robert J.Beichner, John W.Jewett Jr, Contributing author Saunders College Publishing.	- Lectures. - Laboratory experiments. - Problems and essay assignments	2hr 2hr 2hr 2hr 2hr 2hr 2hr 2hr 2hr 2hr 2hr 2hr 2hr	1-Alternating- current generator. 2-The Sine-Wave. 3-Inductance 4-Inductance reactance and circuits. 5-Capacitance. 6-Capacitive reactance and circuits. 7-Alternating- current circuits. 8-Power in A.C circuits. 9-Phase of alternating current. 10-Resonances. 11-Filters. 12-Complex number for A.C. circuit. 13-A.C bridges.	The students have to recognize the following: 1- Elements of A.C circuit structure (Inductance, reactance and capacitance) 2- Phases and resonance in A.C. circuits. 3- Basic devices as filters, bridges and transformers.	

COURSE SYLLABUS محتويات المقرر

الفصل الدراسي الثاني.

PP7& P222	222 ف 7 أ ف	<u>COURSE NO.</u>	1. رقم المقرر :
Bio physics	فيزياء حيوية	<u>COURSE TITLE</u>	2. عنوان المقرر :
2 h lect./w + 1h lab./w		<u>NUMBER OF HOURS</u>	3. عدد الساعات :
2 nd year , 2 nd Semester (Physics & Chemistry sections)		<u>YEAR & SEMISTER</u>	4. الفرقة والفصل الدراسي:
+ شعبة التعليم الابتدائي علوم		<u>COURSE DESCRIPTION</u>	5. وصف المقرر :
The course introduces the study of the fundamentals of application of physics in biological systems.			
		<u>COURSE GOALS:</u>	6 . أهداف المقرر
* Developing a clear understanding of the basis concepts of human body physics			
<u>7. متطلبات السابقة أو المتوازية : PREREQUISITES AND COREQUISITES</u>			

م	مخرجات التعلم Intended Learning Outcomes	الموضوع Topic	الزمن المتوقع لكل موضوع	الأنشطة والواجبات والتعينات Activities, Tasks and Assignment	المصادر التعليمية Educational Resources	التقويم Assessment
	The student have to recognize the following: 1- What are biophysics- biophysical systems? 2- Biophysics techniques. 3- Bioelectric potentials and its applications.	1- Introduction. 2- Electrode technique. 3- Fundamentals of electrochemistry 4- Resting potential. 5- Action potential 6- Surface potentials: - The electro/myogram (E.M.G). - The electro/cardiogram (E.C.G). - The electro/enceptdgram (E.E.G).	2hr 2hr 2hr 2hr 2hr 4hr	- Lectures. - Laboratory experiments - Problems assignments and essay	- Computer simulation programs and slides. - Transparences. - Manual of solved problems (answer and solutions) - Text books. 1- Principles of biophysics, Fadel M.Ali, 2003 2- Biophysics. An introduction, Rodney Cottenill, 2003, John Wiley& Sons LTD	- Semester activities including class room interactions and Quizzes. - Mid-term exam. - Oral Exam. - Lab performance evaluation. - Final Exam.

م	مخرجات التعلم Intended Learning Outcomes	الموضوع Topic	الزمن المتوقع لكل موضوع	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	المصادر التعليمية Educational Resources	التقويم Assessment
	4- Physical properties of living cells with emphasis on passive electrical properties	- The electro/retinogram (E.R.G). 7- Physical properties of living cells. - Dielectric properties, the passive electrical properties.	6hr			
	5- Biomechanics of living cells, forces, in the body Physics of some parts of human body.	8- Bio mechanics. - Mechanical properties of living cells. - Forces on and in the body. - Physics of the skeleton.	6hr			

COURSE SYLLABUS محتويات المقرر
الفصل الدراسي الثاني

٥

P 223

Physical Optics
2 h lect./w + 2h lab./w
2nd year, 2nd semester

بصريات فيزيائية

COURSE NO.

1. رقم المقرر :

COURSE TITLE

2. عنوان المقرر :

NUMBER OF HOURS

3. عدد الساعات :

YEAR & SEMISTER

4. الفرقة والفصل الدراسي :-

COURSE DESCRIPTION

5. وصف المقرر :

The course introduces the students to the fundamentals of Physical optics: Interference, Diffraction and Polarization

COURSE GOALS:

6 . أهداف المقرر

* The students have to know how light interfere, diffract and polarized together with applications of these phenomena

PREREQUESITES AND COREQUISITES : المتطلبات السابقة أو المتوازية :

التقويم Assessment	المصادر التعليمية Educational Resources	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	الزمن المتوقع لكل موضوع	الموضوع Topic	مخرجات التعلم Intended Learning Outcomes	م
- Semester activities including class room interactions and Quizzes. - Mid-term exam - Lab performance evaluation. - Oral exam. - Final exam.	- Computer simulation programs and slides. - Transparencies. - Manual of solved problems (answer and solutions) - Text Books 1- "Fundamentals of Optics", F.A. Jenkins and H.E. White 2- " Fundamentals of Physics", D. Halliday and R. Resnik. 3- Principles of Optics" Max Born and E. Wolf 4-" Geometrical optis H.T. Flint 5- التداخل في الضوء والاطياف أ.د. نايل بركات ، أ.د. أحمد أمين حمزة	- Lectures. - Laboratory experiments. - Problems and essay assignments	2hr 2hr 2hr 2hr 2hr 2hr 2hr 2hr 2hr 2hr 2hr 2hr 2hr	1- Interference 2- Young's experiment. 3- Phase change due to reflection 4- Interferometers and its application 5- Newton rings 6- Interferometers and its application. 7- Diffraction through different slits. 8- Fraunhofer diffraction. 9- Diffraction grating. 10-Resolving power of diffraction grating. 11-Polarization of light. 12-How to obtain a polarized light. 13-Polarization by reflection and disperation. 14-Optical activity polarometer.	The students have to recognize the following: 1- concept of light interference and phase change due to its reflection 2- Experiments that display interference phenomenon. 3- Equipment used in the study of interferometry. 4- Concept of light diffraction. 5- Concept of light polarization 6- Different types of polarization 7-Optical activity and its measurements	

COURSE SYLLABUS المحتويات المقرر

الفصل الدراسي الأول

5

P 311

Atomic Physics

2 h lect./w & 3 h lab / w.

3rd year, 1st semester

COURSE NO.

١. رقم المقرر :

فيزياء ذرية وأطياف

COURSE TITLE

٢. عنوان المقرر :

NUMBER OF HOURS

٣. عدد الساعات :

YEAR & SEMISTER

٤. الفرقة والفصل الدراسي :-

COURSE DESCRIPTION

٥. وصف المقرر :

The course introduces the students to the fundamental of atomic physics, Building a solid foundation for follow on courses in nuclear and solid state physics.

COURSE GOALS:

٦. أهداف المقرر

* The student has to know:

1- The historical development of atomic physics.

2- Structure of the atom and generation of X-rays. Effect of external magnetic fields on atoms

٧. متطلبات السابقة أو المتوازية : PREREQUISITES AND COREQUISITES

٣٤

م	مخرجات التعلم Intended Learning Outcomes	الموضوع Topic	الزمن المتوقع لكل موضوع	الأنشطة والواجبات والتعينات Activities, Tasks and Assignment	المصادر التعليمية Educational Resources	التقويم Assessment
	The Student have to recognize the following: 1- The historical development of the atomic structure including Bohr's model Sommerfield model and Vector model.	1- Historical background. 2- Thomson's model. 3- Rutherford's Model. 4- Line spectrum- Energy levels. 5- Hydrogen atom spectrum. 6- Bohr's atomic model. 7- Frank- Hertz Exp. 8- Sommerfield Model. 9- Vector model. 10- Many electron atoms. 11- Stern-Gerlach experiment. 12- Pauli Exclusion Principle. 13- Periodic table.	4hr 2hr 2hr 4hr 2hr 2hr	- Lectures. - Laboratory experiments. - Problems and essay assignments	- Computer simulation programs and slides. - Transparencies. - Manual of solved problems (answer and solutions) - Text books: 1- An Introduction to atomic and nuclear physics by H.Semat. 2- Advanced Physics by Kieth Gibbs. 3- Modern Physics by Weidner and sells	- Semester activities including classroom interactions and quizzes. - Mid-term exam. - Oral exam. - Final exam.

التقويم Assessment	المصادر التعليمية Educational Resources	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	الزمن المتوقع لكل موضوع	الموضوع Topic	مخرجات التعلم Intended Learning Outcomes	م
			4hr 2hr 2hr 2hr	14- X-rays. 15- X-rays production. 16- X-rays line spectra. 17- X-rays continuous spectra. 18- Electronic configuration. 19- The normal Zeeman Effect. 20- Anomalous Zeeman Effect. 21- Doublet fine structure 22- Selection rules for possible transition.	2- X-ray production and properties. 3- Normal and Anomalous Zeeman effects	

COURSE SYLLABUS محتويات المقرر

الفصل الدراسي الأول

P 312

Electron optics

2 h lect./w + 3h lab./w

3rd year, 1st semester

بصريات إلكترونية

COURSE NO.

COURSE TITLE

NUMBER OF HOURS

YEAR & SEMISTER

COURSE DESCRIPTION

1. رقم المقرر :

2. عنوان المقرر :

3. عدد الساعات :

4. الفرقة والفصل الدراسي :

5. وصف المقرر :

The course introduces the students to the principles of electron emission, motion in electromagnetic field, construction of electric and magnetic lenses and electron microscope

COURSE GOALS:

6. أهداف المقرر

- * Developing and understanding the applications of electron motion in E.M. field namely construction of electronic lenses and electron microscope.

PREREQUESITES AND COREQUISITES : متطلبات السابقة أو المتوازية :

م	مخرجات التعلم Intended Learning Outcomes	الموضوع Topic	الزمن المتوقع لكل موضوع	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	المصادر التعليمية Educational Resources	التقويم Assessment
	The Student have to recognize the following: 1- Electron motion in E.M field. 2- The structure of electronic lenses of different configuration and compansion with optical lenses. 3- Basic construction and performance transmission and scanning electron microscope and applications.	1- Introduction to electron optics. 2- Motion of electrons in uniform: electrostatic and magnetic fields. 3- Types and design of electron lenses. 4- Physical similarity of light and electron lenses. 5- Types of electron microscope and applications.	4hr 6hr 6hr 2hr 8hr	- Lectures. - Laboratory experiments. - Problems and essay assignments	- Computer simulation programs and slides. - Transparencies. - Manual of solved problems (answer and solutions) - Text books: 1- Hawkes, P.W. (1972)" Electron optics and electron microscopy (Taylor limited)." 2- Zowrykin, v. (1957) Electron optics and electron microscope. John Wiley and Sons. Inc 3- Hawkes PW, Kasper (1996): Principle of electron optics. Academic press, London and San Diego	- Semester activities including classroom interactions and quizzes. - Mid-term exam. - Oral exam. - Final exam.

COURSE SYLLABUS محتويات المقرر

الفصل الدراسي الأول

١٥

P 313

Quantum Mechanics (1)

2 h lect./w

3rd year, 1st semester

COURSE NO.

COURSE TITLE

NUMBER OF HOURS

YEAR & SEMISTER

COURSE DESCRIPTION

1. رقم المقرر :

2. عنوان المقرر :

3. عدد الساعات :

4. الفرقة والفصل الدراسي :

5. وصف المقرر :

The course introduces the students to the principles of classical mechanics as an introduction to study quantum mechanics and its applications.

COURSE GOALS:

6 : أهداف المقرر

* Building a solid foundation for follow – on courses in atomic, nuclear and solid state physics.

7. متطلبات السابقة أو المتوازية : PREREQUESITES AND COREQUISITES

م	مخرجات التعلم Intended Learning Outcomes	الموضوع Topic	الزمن المتوقع لكل موضوع	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	المصادر التعليمية Educational Resources	التقويم Assessment
	The Student have to recognize the following: 1- Lagrange's & Hamiltons formulations of mechanics. 2- Black body radiation and Planck's hypothesis. 3- Photo- electric effect, De-Broglie hypothesis and Heisenberg uncertainty principle. 4- Postulates of Q. Mech. And the solution of one dimensional	I- <u>Classical Mechanics</u> : 1 - LaGrange's & Hamilton's formulation of Mechanics. 2 - System of particles & Rigid bodies. II - <u>Quantum Mechanics</u> : 3- Black body radiation & Plank's Hypothesis. 4- Photo electric effect and Einstien photon theory. 5- wave properties of particles (De- Broglie hypothesis) 6- Electron diffraction. 7- Heisenberg uncertainty principle. 8- Postulates of Q.Mech. 9- Hermiton operators – space functions- Eigen	2hr 4hr 2hr 2hr 2hr 4hr	- Lectures. - Problems and essay assignments	- Computer simulation programs and slides. - Transparences. - Manual of solved problems (answer and solutions) - Text books: 1- Quantum physics by Eisberg and Resnick. 2- Modern physics and quantum mechanics by Anderson	- Semester activities including classroom interactions and quizzes. - Mid-term exam. - Oral exam. - Final exam.

التقويم Assessment	المصادر التعليمية Educational Resources	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	الزمن المتوقع لكل موضوع	الموضوع Topic	مخرجات التعلم Intended Learning Outcomes	م
			2hr 2hr 2hr 2hr 2hr	functions, eigen values – operators. III- Time independent Sch.eq. 10- One dimensional problems: A- Free particles. B- Free particle in a box. C- Potential step (jump) D- Potential well. E- Tunnel effect.	problems for different potential configuration	

COURSE SYLLABUS محتويات المقرر

الفصل الدراسي الأول

P 314

special theory of relativity

2 h lect./w + 1h tut./w

3rd year, 1st semester

COURSE NO.

النظرية النسبية الخاصة

COURSE TITLE

NUMBER OF HOURS

YEAR & SEMISTER

COURSE DESCRIPTION

١. رقم المقرر :

٢. عنوان المقرر :

٣. عدد الساعات :

٤. الفرقة والفصل الدراسي :

٥. وصف المقرر :

The course introduces the students to the principles of the special theory of relativity

COURSE GOALS:

٦. أهداف المقرر

* Building a solid foundation for electro dynamics and advanced courses in atomic and nuclear physics.

٧. متطلبات السابقة أو المتوازية : PREREQUISITES AND COREQUISITES

م	مخرجات التعلم Intended Learning Outcomes	الموضوع Topic	الزمن المتوقع لكل موضوع	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	المصادر التعليمية Educational Resources	التقويم Assessment
	The Student have to recognize the following: 1- The difference between Galilean and Lorentz transformations. 2- Length contraction time dilation and the transformation of velocities and accelerations. 3- Relativistic mass and mass energy equivalent.	1- Frame of references. 2- Galilean transformations. 3- Michelson & Morley Exp. 4- Lorentz trans. 5- Length contraction and time dilation. 6- Transformation of velocity and acceleration. 7- Relativistic mass. 8- Equivalence of mass and energy. 9- Total energy and potential energy in relativistic forms 10- Applications of relativistic dynamics.	2hr 2hr 2hr 4hr 4hr 4hr 4hr 4hr	- Lectures. - Problems and essay assignments	- Computer simulation programs and slides. - Transparencies. - Manual of solved problems (answer and solutions) - Text books: 1- Relativity and Early Quantum theory by Robert Resnik	- Semester activities including class room interactions and Quizzes. - Mid-term exam - Oral exam. - Final exam.

COURSE SYLLABUS محتويات المقرر

الفصل الدراسي الأول

٥

P 315

Statistical Physics

2 h lect./w + 1h tut./w

3rd year, 1st semester

COURSE NO.

COURSE TITLE

NUMBER OF HOURS

YEAR & SEMISTER

COURSE DESCRIPTION

The course introduces the students to the most fundamental physical concepts of "Statistical Physics" and to familiarize student with the basic notations of probability theory.

COURSE GOALS:

- * Developing a clear understanding of the basic concepts in statistical physics as a basic prerequisite to study solid state physics.

PREREQUESITES AND COREQUISITES : متطلبات السابقة أو المتوازية :

م	مخرجات التعلم Intended Learning Outcomes	الموضوع Topic	الزمن المتوقع لكل موضوع	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	المصادر التعليمية Educational Resources	التقويم Assessment
	The Student have to recognize the following: 1- How a system of many particles is described in statistical terms and understanding of macroscopic systems based on counting its accessible states. 2- Understanding basic concepts as entropy, absolute temperature and different probability distributions.	1- Energy states and energy levels. 2- Macro states and microstates. 3- Thermodynamic probability: - MB – Statistics - BE – Statistics. - FD- Statistics. 4- The Statistical interpretation of entropy. 5- MB and classical distribution functions Equipartition principle.	2hr 3hr 3hr 3hr 3hr 2hr 2hr	- Lectures. - Problems and essay assignments	- Computer simulation programs and slides. - Transparences. - Manual of solved problems (answer and solutions). - Text Book	- Semester activities including class room interactions and quizzes. - Mid-term exam. - Oral exam. - Final exam.

م	مخرجات التعلم Intended Learning Outcomes	الموضوع Topic	الزمن المتوقع لكل موضوع	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	المصادر التعليمية Educational Resources	التقويم Assessment
	3- Discussing how one relates atomic concepts to macroscopic measurements.	6- The partition function. 7- Application to gases and other systems	2hr 6hr			

COURSE SYLLABUS المحتويات المقرر

الفصل الدراسي الثاني

P 321

Nuclear Physics (1)

2 h lect./w + 2h lab/ w

3rd year, 2nd semester

COURSE NO.

COURSE TITLE

NUMBER OF HOURS

YEAR & SEMISTER

COURSE DESCRIPTION

١. رقم المقرر :

٢. عنوان المقرر :

٣. عدد الساعات :

٤. الفرقة والفصل الدراسي :

٥. وصف المقرر :

The course introduces the students to principles of nuclear physics by studying the constituents of the nucleus and the natural radioactivity

COURSE GOALS:

٦. أهداف المقرر

- 1- Developing a clear understanding of the basic concepts in nuclear physics.
- 2- Building a solid foundation for advanced nuclear physics course

٧. متطلبات السابقة أو المتوازية : PREREQUISITES AND COREQUISITES

م	مخرجات التعلم Intended Learning Outcomes	الموضوع Topic	الزمن المتوقع لكل موضوع	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	المصادر التعليمية Educational Resources	التقويم Assessment
	The Student have to recognize the following: 1- The early researches for measuring nuclear dimensions, charges and masses. 2- The constituents of the nucleus.	1- Rutherford theory of alpha scattering. The classical properties of the nuclear radius, charge and mass and how to determine each of them. 2- The constituents of the nucleus: electron-proton theory, proton- neutron model. 3- Applying the principles of wave mechanics to look for the presence of electrons and protons inside the nucleus. 4- Discovery of neutrons- quark theory. The nuclear	8hr 10hr	- Lectures. - Laboratory experiments. - Problems and essay assignments	- Computer simulation programs and slides. - Transparencies. - Manual of solved problems (answer and solutions) - Text Books 1- المبادئ الأساسية للفيزياء الذرية. ريتشارد ف. همفريز و روبرت بيرنجر- ترجمة د. محمود أمين عمر وآخرين. دار المعارف بمصر 1962. 2- مقدمة في الفيزياء الذرية والنوية - هنري سيمات ترجمة د/ مصطفى كامل ود. سيد رمضان هدارة - مكتبة النهضة المصرية- 1967. والطبعة الانجليزية الأصلية لنفس	- Semester activities including class room interactions and Quizzes. - Mid-term exam - Lab performance evaluation. - Oral exam. - Final exam.

م	مخرجات التعلم Intended Learning Outcomes	الموضوع Topic	الزمن المتوقع لكل موضوع	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	المصادر التعليمية Educational Resources	التقويم Assessment
	4- Nuclear structure and nuclear models.	4- Nuclear structure and nuclear models: Liquid drop model and the shell model as examples.	4hr			
	5- Neutron and reactor physics.	5- Neutron and reactor physics: Production of neutrons- Neutron reactions and detection- Fission reactions- Principles of reactors: (Types- Fuels- moderators- coolants). Infinite amplification coefficients- critical volume- Examples- Fussion reactions.	4hr.			
	6- Accelerators	6- Accelerators: Importance- Types- Uses 7- Introduction to elementary particles.	4hr			

م	مخرجات التعلم Intended Learning Outcomes	الموضوع Topic	الزمن المتوقع لكل موضوع	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	المصادر التعليمية Educational Resources	التقويم Assessment
	3- The natural radioactivity and the laws governing it	binding energy, nuclear fission and fusion reactions. 5- The natural radioactivity theory, the laws of radioactive disintegrations. Specific radioactivity and units of radioactivity-branching-Successive radioactivity-radioactive equilibrium	8hr		المؤلف. 3- الفيزياء النووية د. محمد شحادة الدغمة ود. على محمد جمعة - اصدار دار العزيز للطباعة والنشر - دبي - الامارات العربية المتحدة - الجزء الأول والثاني 1997. 1- Kaplan, I (1969) Nuclear physics 2 nd edition, Adison Wesley publishing company, California U.S.A.	

COURSE SYLLABUS المحتويات المقرر

الفصل الدراسي الثاني

(٢)

P 322

Solid State (1)

2 h lect./w + 2h lab./w

3rd year, 2nd semester

COURSE NO.

COURSE TITLE فيزياء جوامد (1)

NUMBER OF HOURS

YEAR & SEMISTER

COURSE DESCRIPTION

1. رقم المقرر :

2. عنوان المقرر :

3. عدد الساعات :

4. الفرقة والفصل الدراسي :

5. وصف المقرر :

The course introduces the students to principles of the structure and physics of material as an introductory course for solid state physics.

COURSE GOALS:

6. أهداف المقرر

* Developing the student's ability to deal with the concepts of crystal structure, thermal properties and magnetic properties of materials.

7. متطلبات السابقة أو المتوازية : PREREQUISITES AND COREQUISITES

التقويم Assessment	المصادر التعليمية Educational Resources	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	الزمن المتوقع لكل موضوع	الموضوع Topic	مخرجات التعلم Intended Learning Outcomes	م
- Semester activities including class room interactions and Quizzes. - Mid-term exam - Lab performance evaluation. - Oral exam. - Final exam.	- Computer simulation programs and slides. - Transparencies. - Manual of solved problems (answer and solutions) - Text Books 1- "Elements of X-ray diffraction", B.D. Cullity, Addison-Wesley pub. Company.Inc. 2- " Introduction to solid state physics", C.Kittel, John Wiley& son,Inc. 3- " Introduction to solids", L.V Azaroff, McGrew- Hill 4- " Structure of Metals", C.S Barrett,	- Lectures. - Laboratory experiments. - Problems and essay assignments	2hr 2hr 4hr 2hr 2hr 2hr 4hr	1- Structure of solids: Crystalline and amorphous. 2- X-ray diffraction. 3- Applications of X-ray crystallography phase identification, crystallite size, residual internal strain and crystallographic texture. 4- Types of crystal imperfections. 5- Interatomic forces and classification of solids. 6- Lattice vibration. 7- Thermal properties: classical Einstein and Debby theories of	The Student have to recognize the following: 1- Structure of solids and applications of X-ray crystallography. 2- Types of Crystal imperfections 3- Lattice vibration and thermal properties of materials.	

م	مخرجات التعلم Intended Learning Outcomes	الموضوع Topic	الزمن المتوقع لكل موضوع	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	المصادر التعليمية Educational Resources	التقويم Assessment
	4- Magnetic properties of materials and types of magnetism.	heat capacity. 8- Thermal conductivity and thermal expansion. 9- Magnetic properties and classical theory. 10- Types of magnetism: Dia-, para- and ferro magnetism	2hr 2hr 4hr		McGraw Hill. 5- " Solid state physics ", A.J. Dekker, MacMillan Press Ltd. 6- " An Introduction to solid state Physics", R.J Elliot and A.F. Gibson. 7- " Theoretical structure Metallurgy", A.H. Cottrell, E.L.B.S.& Edward Arnold pub.Ltd.	

COURSE SYLLABUS محتويات المقرر

الفصل الدراسي الثاني

P 323

Quantum physics (2)

2 h lect./w + 1h tut./w

3rd year, 2nd semester

COURSE NO.

COURSE TITLE

NUMBER OF HOURS

YEAR & SEMISTER

COURSE DESCRIPTION

1. رقم المقرر :

2. عنوان المقرر :

3. عدد الساعات :

4. الفرقة والفصل الدراسي :-

5. وصف المقرر :

The course introduces the students to principles of quantum mechanics including 3 dimensional problems as a continuation to quantum physics (1)

COURSE GOALS:

6 . أهداف المقرر

- * Developing a clear understanding of the basic concepts of quantum mechanics by studying a real system qualitatively and quantitatively as hydrogen atom

PREREQUISITES AND COREQUISITES : متطلبات السابقة أو المتوازية :

Quantum Physics (1)

التقويم Assessment	المصادر التعليمية Educational Resources	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	الزمن المتوقع لكل موضوع	الموضوع Topic	مخرجات التعلم Intended Learning Outcomes	م
- Semester activities including class room interactions and quizzes. - Mid-term exam. - Oral exam. - Final exam.	- Computer simulation programs and slides. - Transparencies. - Manual of solved problems (answer and solutions) - Text Books 1- " Quantum Physics" by M.S.Rogalsk & S.B. Palmer. 2- Understanding Quantum Mechanics, by Morrison 3- الفيزياء الكمية مقرر بيركلي في الفيزياء تأليف أ.ديكمان، ترجمة أ.د. خليل محمد إبراهيم عبده، أ.د. محمد عبد الله السمرى، مراجعة أ.د. محمد عبد المقصود النادى.	- Lectures. - Problems and essay assignments	4hr 4hr 8hr 2hr 2hr 2hr 4hr	1-Free particle in 3 dimensions 2-Free particle in 3 dimensional box. 3-Three dimensional harmonic oscillator. 4-Central potential and Quantum Mechanical treatment of Hydrogen atom. 5-Quantum numbers. 6-Angular momentum for electron. 7-The spinning of electron Periodic table. 8-Commutation relations.	The Student has to solve 3 dimensional problems using principles of quantum mechanics in the following topics. - Free particle-particle in a box. - Simple harmonic oscillator as a basic system for the description of vibrating molecules atoms and nuclei. - Coulomb field as a prototype potential for atomic structure. - Angular momentum and commutation relations	

COURSE SYLLABUS محتويات المقرر

الفصل الدراسي الثاني

2

P 324

Computational Physics

2 h lect./w + 1h tut./w

3rd year, 2nd semester

COURSE NO.

فيزياء حسابية COURSE TITLE

NUMBER OF HOURS

YEAR & SEMISTER

COURSE DESCRIPTION

1. رقم المقرر :

2. عنوان المقرر :

3. عدد الساعات :

4. الفرقة والفصل الدراسي :

5. وصف المقرر :

The course introduces the students to principles of numerical techniques and its applications in physics problems.

COURSE GOALS:

6. أهداف المقرر

* Developing the student's problem solving, and skills needed to find numerical solutions and graphs for physics problems.

7. متطلبات السابقة أو المتوازية : PREREQUISITES AND COREQUISITES

التقويم Assessment	المصادر التعليمية Educational Resources	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	الزمن المتوقع لكل موضوع	الموضوع Topic	مخرجات التعلم Intended Learning Outcomes	م
- Semester activities including class room interactions and Quizzes. - Mid-term exam - Lab performance evaluation. - Oral exam. - Final exam.	- Computer simulation programs and slides. - Transparencies. - Manual of solved problems (answer and solutions) - Text Books 1- Applied numerical analysis by Gerald Addison- Wesley. 2- Elementary numerical analysis by Samuel D.Conte & Carlde Boor (McGrew Hill). 3- An Introduction to numerical analysis by Kendall E.A Liknson (Wiley)	- Lectures. - Problems and essay assignments. - Computer Simulations	4hr 6hr 4hr 6hr 6hr	1- Computational errors- Floating point- Root finding- Bisection method- Newton's method. 2- Polynomial approximation. Least square method. 3- Numerical differentiation and integration methods. 4- Numerical solution of ordinary differential equations. 5- Numerical solution for a system of linear differential equations.	The students have to recognize the following: 1- Computational errors in physical measurements. 2- Using least square fitting technique and making graphs. 3- Numerical methods of differentiation and integration. 4- Using numerical analysis for solving differential equations.	

COURSE SYLLABUS المحتويات المقرر

الفصل الدراسي الثاني

P 325

Use of computer in physics

2 h lect./w + 2h lab./w

3rd year, 2nd semester

استخدام الحاسب في الفيزياء

COURSE NO.

COURSE TITLE

NUMBER OF HOURS

YEAR & SEMISTER

COURSE DESCRIPTION

1. رقم المقرر :

2. عنوان المقرر :

3. عدد الساعات :

4. الفرقة والفصل الدراسي :

5. وصف المقرر : -

The course introduces the students to principles of transducers, their applications in computerizing physics data collected in the laboratory and analysis of these data

COURSE GOALS:

6. أهداف المقرر

* Developing the student skills in using interfaces to computers, the graphical and statistical analysis of the collected data.

PREREQUISITES AND COREQUISITES : متطلبات السابقة أو المتوازية :

التقويم Assessment	المصادر التعليمية Educational Resources	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	الزمن المتوقع لكل موضوع	الموضوع Topic	مخرجات التعلم Intended Learning Outcomes	م
- Semester activities including class room interactions and Quizzes. - Mid-term exam - Lab performance evaluation. - Oral exam. - Final exam.	- Computer simulation programs and slides. - Transparences. - Manual of solved problems (answer and solutions) - Text Books 1-"Microprocessor and Microcomputers" by D.D.Givone and R.P.Roeser. 2- " Electronics" by J.M Calvert and J.Mc Cauland, Joha Willy and sons.	- Lectures. - Laboratory experiments. - Problems and essay assignments	2hr 6hr 6hr 6hr 6hr	1- Introduction. 2- Transducers - Heat/ Conversion into voltage. -Light/ Kinetical displacement. 3- Types of interfaces to computers. - General and special purpose interface cards. 4- Use of transducers and interface cards as input/ output controllers. 5- Use of computers programs in graphical and statistical analysis of the collected data.	The students have to recognize the following: 1- Different types of transducers. 2- General and special purpose interface cards. 3- The use of transducers and interface cards as input / output controllers. 4- Using computer programs in graphical and statistical analysis of the collected data.	

COURSE SYLLABUS المحتويات المقر

الفصل الدراسي الأول

P411

Nuclear Physics(2)

2 h lect./w + 2h lab./w

4th year, 1st semester

COURSE NO.

COURSE TITLE

NUMBER OF HOURS

YEAR & SEMISTER

COURSE DESCRIPTION

1. رقم المقرر :

2. عنوان المقرر :

3. عدد الساعات :

4. الفرقة والفصل الدراسي :

5. وصف المقرر :

The course introduces the students to nuclear forces, reactors and accelerators.

COURSE GOALS:

6. أهداف المقرر

- * The students have to know structure of the nucleus in terms of nuclear forces and its applications in both nuclear reactors and accelerators.

PREREQUISITES AND COREQUISITES : متطلبات السابقة أو المتوازية :

Nuclear Physics (1)

التقويم Assessment	المصادر التعليمية Educational Resources	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	الزمن المتوقع لكل موضوع	الموضوع Topic	مخرجات التعلم Intended Learning Outcomes	م
- Semester activities including class interactions and quizzes. - Mid-term exam - Lab performance evaluation. - Oral exam. - Final exam.	- Computer simulation programs and slides. - Transparences. - Manual of solved problems (answer and solutions) - Text books: 1- Nuclear physics by I.kaplan (Adison Wesley). 2- Nuclear physics and Nuclear reactors by A.Klimov (Mir publishors). 3- α, β and ray spectroscopy (Vol I) by K.Sigban (North Holand publishing)	- Lectures. - Laboratory experiments. - Problems and essay assignments. - Visits to nuclear research Centers and reactor.	4hr	1- Electric and Magnetic properties of the nucleus. The quantum numbers of the nucleon inside the nucleus. The nuclear energy levels. The Vector coupling. The magnetic moment of the nucleus. The parity.	The student have to recognize the following: 1- Electric and magnetic properties of the nucleus and the concept of parity.	
			4hr	2- The nuclear forces, Schrodinger's wave equation of duetron.	2- Nature of nuclear forces with an application to the simplist nuclear system.	
			6hr	3- Theories and mechanisms of Alpha, Beta and Gamma decay.	3- Mechanisms of Alpha, Beta and Gamma decays and the theories governing these processes.	

م	مخرجات التعلم Intended Learning Outcomes	الموضوع Topic	الزمن المتوقع لكل موضوع	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	المصادر التعليمية Educational Resources	التقويم Assessment
	4- Nuclear structure and nuclear models.	4- Nuclear structure and nuclear models: Liquid drop model and the shell model as examples.	4hr			
	5- Neutron and reactor physics.	5- Neutron and reactor physics: Production of neutrons- Neutron reactions and detection- Fission reactions- Principles of reactors: (Types- Fuels- moderators- coolants). Infinite amplification coefficients- critical volume- Examples- Fussion reactions.	4hr.			
	6- Accelerators	6- Accelerators: Importance- Types- Uses	4hr			

COURSE SYLLABUS المحتويات المقرر

الفصل الدراسي الأول

(د)

P412

Solid State (2)

2 h lect./w + 2h lab./w

4th year, 1st semester

COURSE NO.

فيزياء جوامد (2)

COURSE TITLE

NUMBER OF HOURS

YEAR & SEMISTER

COURSE DESCRIPTION

COURSE GOALS:

* Building solid foundation forthcoming developments in the domain of solid state.

PREREQUESITES AND COREQUISITES : متطلبات السابقة أو المتوازية :

Solid State (1)

1. رقم المقرر :

2. عنوان المقرر :

3. عدد الساعات :

4. الفرقة والفصل الدراسي :-

5. وصف المقرر :

6 . أهداف المقرر

م	مخرجات التعلم Intended Learning Outcomes	الموضوع Topic	الزمن المتوقع لكل موضوع	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	المصادر التعليمية Educational Resources	التقويم Assessment
	The student have to recognize the following: 1- Electrical properties of solids. 2- Band theory of solids. 3- Types of junctions.	1- Electric properties of solids and free electron theory 2- Structure dependence of resistivity and Matthiessen rule. 3- Wiedmann- Franz law. 4- Bloch theorem. 5- Koring Penny model. 6- Band theory: metals, semiconductors (intrinsic and extrinsic) and insulators. 7- Temperature dependence of resistivity. 8- Motion of electron in magnetic field and Hall effect. 9- Junction properties metal-metal, metal-	2hr 1hr 1hr 1hr 1hr 4hr 2hr 2hr 2hr	- Lectures. - Laboratory experiments. - Problems and essay assignments.	- Computer simulation programs and slides. - Transparences. - Manual of solved problems (answer and solutions) - Text books: 1- "Introduction to solid state physics", C.Kittel, John Wiley & Sons, Inc. 2- "Introduction to Solids", L.V. Azaroff, McGrew – Hill. 3- "An Introduction to solid state physics", R.J.Elliot and A.F.Gibson, Macmillan Press, LTD.	- Semester activities including class interactions and quizzes. - Mid-term exam - Lab performance evaluation. - Oral exam. - Final exam.

التقويم Assessment	المصادر التعليمية Educational Resources	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	الزمن المتوقع لكل موضوع	الموضوع Topic	مخرجات التعلم Intended Learning Outcomes	م
	4- Elementary Solid State Physics. By M.Ali Omar Revised Printing Addison Wesley Longman 1993. 5- The Physics of Solids By R.turton, Oxford Univ.Press.2000		2hr 2hr. 1hr 1hr 2hr 2hr	semiconductor and P-N junctions. 10- Dielectrics and Mechanisms of polarization. 11- Piezo-, pyro- and ferro-electric materials. 12- Optical properties of semiconductors. 13- Optical properties of insulators. 14- Color of crystals. 15- Photoconductivity	4- Dielectrics and their types. 5- Optical properties of solids.	

COURSE SYLLABUS محتويات المقرر

الفصل الدراسي الثاني

P413

Laser physics

2 h lect./w + 2h lab./w

4th year, 2nd semester

COURSE NO.

فيزياء الليزر

COURSE TITLE

NUMBER OF HOURS

YEAR & SEMISTER

COURSE DESCRIPTION

1. رقم المقرر :

2. عنوان المقرر :

3. عدد الساعات :

4. الفرقة والفصل الدراسي :

5. وصف المقرر :

The course introduces the students to fundamentals, operation and applications of laser together with fibre optics.

COURSE GOALS:

6. أهداف المقرر

- * The students have to know fundamentals of quantum transitions in atomic systems, basis of coherence, Lase production, interactions and applications, optical fibers and optical processing of images.

PREREQUESITES AND COREQUISITES : متطلبات السابقة أو المتوازية :

التقويم Assessment	المصادر التعليمية Educational Resources	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	الزمن المتوقع لكل موضوع	الموضوع Topic	مخرجات التعلم Intended Learning Outcomes	م
- Semester activities including class interactions and quizzes. - Mid-term exam - Lab performance evaluation. - Oral exam. - Final exam.	- Computer simulation programs and slides. - Transparencies. - Manual of solved problems (answer and solutions) - Text books: 1- "Principles of Optics", M.Born and E.Wolf, Cambridge Univ.Press, England. 2- "Fundamentals of Photonics", B.E.A. Saleh, John Wiley & Sons, Inc. 3- "Laser Fundamentals", W.T.Silfvast, Cambridge Univ.Press,	- Lectures. - Laboratory experiments. - Problems and essay assignments.	2hr	1- Introduction:.	The student have to recognize the following: 1- Monochromaticity, directionality and brightness. 2- Laser production and properties. 3- Types of laser and its applications.	
			2hr	2- Quantum transition in an atomic system.		
			2hr	3- Properties of laser beam.		
			2hr	4- Basics of coherence theory.		
			2hr	5- Monochromaticity, directionality and brightness		
			2hr	6- Types of laser beam.		
			2hr	7- Optical feedback and laser resonator.		
			4hr	8- Laser applications: speckles, holography and materials interactions.		
			4hr	9- Fibre optics: materials and manufacture.		
			2hr	10- Integrated optics.		
			2hr	11- Optical processing of images (endscopy, wave carrier).		
			2hr	12- Non linear optics		

النقويم Assessment	المصادر التعليمية Educational Resources	الأنشطة والتوجيهات والتمارين Activities, Tasks and Assignment	الزمن المتوقع لكل موضوع	الموضوع Topic	مخرجات التعلم Intended Learning Outcomes	م
	New York. 4- "Optical Holography", P.Hariharam, Cambridge Univ.Press, New York				4- Non linear optics	

COURSE SYLLABUS محتويات المقرر

الفصل الدراسي الأول

P414	<u>COURSE NO.</u>	1. رقم المقرر :
Electronic Circuits	<u>COURSE TITLE</u>	2. عنوان المقرر :
2 h lect./w + 2h lab./w	<u>NUMBER OF HOURS</u>	3. عدد الساعات :
4 th year, 1 st semester (Physics section)	<u>YEAR & SEMISTER</u>	4. الفرقة والفصل الدراسي :
3 rd year, 2 nd Semester (Chemistry section)		

COURSE DESCRIPTION 5. وصف المقرر :

The course introduces the students to the basic elements of electronic circuits namely electronic diodes, transistors and integrated circuits.

COURSE GOALS: 6. أهداف المقرر

- * The students have to know characteristics of basic semiconductor devices and their role in the construction of electronic circuits with emphasis on integrated circuits.

PREREQUISITES AND COREQUISITES 7. متطلبات السابقة أو المتوازية :

التقويم Assessment	المصادر التعليمية Educational Resources	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	الزمن المتوقع لكل موضوع	الموضوع Topic	مخرجات التعلم Intended Learning Outcomes	م
Semester activities including class - Interactions and Quizzes. - Mid-term exam - Lab performance evaluation. - Final exam.	- Computer simulation programs and slides. - Transparencies. - Manual of solved problems (answer and solutions) - Text books: 1- "Electronics" by J.M Cavert and M.A.H. McCanland, John Willy & Sons. 2- "Integrated Electronics" by Hillman and Halkiac, John Willy	- Lectures. - Laboratory experiments. - Problems and essay assignments.	10hr	1- Semi conductor devices: (a) Electronic diodes: p.n junction- light emmiting diode, photo diode, zenerdiode, laser diode and tunneling diode (b) Transistors and characteristics P-N-P and N-P-N, field effect transistor.	1- The Types of semi conductor devices and their characteristics.	
			10hr	2- Circuits (a) Diode circuits; Rectifiers, stabilizers, photodetectors, oscillators. (b) Transistor as: amplifiers, multivibrators, oscillators and summing amplifier.	2- The use of semiconductor devices in some electronic circuits.	
			10hr	3- Integrated circuits (IC); timer and amplifier.	3- Principles of integrated circuits.	

محتويات المقرر
COURSE SYLLABUS
الفصل الدراسي الأول

١٤

P415

Plasma Physics

2 h lect./w + 1h tut./w

4th year, 1st semester

COURSE NO.

فيزياء البلازما COURSE TITLE

NUMBER OF HOURS

YEAR & SEMISTER

COURSE DESCRIPTION

1. رقم المقرر :

2. عنوان المقرر :

3. عدد الساعات :

4. الفرقة والفصل الدراسي :

5. وصف المقرر :

The course introduces the students to basic principle of plasma physics with emphasis of plasma sources, fusion research and space plasma physics.

COURSE GOALS:

6 : أهداف المقرر

- * Developing the student's skills and creative thought needed to meet new trends in plasma, fusion research and space plasma physics.

7. متطلبات السابقة أو المتوازية : PREREQUISITES AND COREQUISITES

التقويم Assessment	المصادر التعليمية Educational Resources	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	الزمن المتوقع لكل موضوع	الموضوع Topic	مخرجات التعلم Intended Learning Outcomes	م
- Semester activities including class interactions and quizzes. - Mid-term exam - Computer lab evaluation. - Oral exam. - Final exam.	- Computer simulation programs and slides. - Transparencies. - Manual of solved problems. - Text Books: 1- "Introduction to Plasma Phys. And Controlled fusion" by F. Chen, Plenum Press. 2- "Introduction to Plasma Theory" by D.R. Nicholson, Wiley publisher. 3- "Waves in dusty space plasmas" by F. Verheest, Kluwer pub. 4- WWW.Plasmas.org 5- WWW.Plasma physics course	- Lectures. - Problem solving and essay assignments.	6hr	1- Characteristics of plasma, the plasma interaction with a magnetic field (magnet-hydropmagnetic fluid), detailed maps of the magnetic field on the sun spots, the ionosphere, the magnetic storms, the aurora belts of energetic radiation above the earth's atmosphere, the direct conversion of the kinetic energy in a gas to electricity, ion jet propulsion and controlled molecular reaction.	The students have to recognize the following: 1- Characteristics of plasma and plasma interaction with the magnetic field of Sun's surface, sun spot, the ionosphere, magnetic storms and the aurora belts.	
			4hr	2- Debye radius, motion	2- Motion of charged	

التقويم Assessment	المصادر التعليمية Educational Resources	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	الزمن المتوقع لكل موضوع	الموضوع Topic	مخرجات التعلم Intended Learning Outcomes	م
			4hr	of charged particles in a uniform magnetic field, Plasma Larmoy frequency and plasma magnetic moment. Motion of charged particles in a non-uniform magnetic field, the magnetic mirror, drift in Toroidal magnetic field, rotating plasma, and pinch effect.	particlec in time dependent magnetic field and toroidal magnetic field.	
			4hr	3- Plasma electron oscillation, ion oscillation and waves, Debye screening, transmission of electro magnetic waves through plasma, hydromagnetic waves / Alfven waves. 4- Lagrange interaction, relaxation time,	3- Transmission of electromagnetic waves through plasma and Alfven waves. 4- Transmission	

التقويم Assessment	المصادر التعليمية Educational Resources	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	الزمن المتوقع لكل موضوع	الموضوع Topic	مخرجات التعلم Intended Learning Outcomes	م
			4hr	electrical conductivity of plasma, thermal conductivity and viscosity of plasma	coefficient and plasma instabilities	
			4hr	5- Bermsstrahlung radiation, radiation emitted by excited ions and atoms, betatron emission from plasma Measurements of currents and voltages of moving plasma. Electron density- Flux density and plasma spectrometry.	5- Plasma radiation and diagnostic techniques.	
				6- Waste production, safty. Introduction to fusion reactors, fusion reactors, self-sustaining reactions, fusion reactors and Tokamak and applications.	6- Nuclear fusion, reactors and Tokamak	

COURSE SYLLABUS محتويات المقرر

الفصل الدراسي الأول

P 416

Electrodynamics

2 h lect./w + 1h tut./w

4th year, 1st semester

COURSE NO.

ديناميكا كهربية COURSE TITLE

NUMBER OF HOURS

YEAR & SEMISTER

COURSE DESCRIPTION

١. رقم المقرر :

٢. عنوان المقرر :

٣. عدد الساعات :

٤. الفرقة والفصل الدراسي :

٥. وصف المقرر :

The course introduces the students to solutions of Maxwell's equation and its applications

COURSE GOALS:

٦. أهداف المقرر

- * The students have to know how emission of electromagnetic radiation together with other physical phenomena could be described in terms of Maxwell's equations

٧. متطلبات السابقة أو المتوازية : PREREQUESITES AND COREQUISITES

التقويم Assessment	المصادر التعليمية Educational Resources	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	الزمن المتوقع لكل موضوع	الموضوع Topic	مخرجات التعلم Intended Learning Outcomes	م
Semester activities including class - Interactions and Quizzes. - Mid-term exam - Lab performance evaluation. - Final exam.	- Computer simulation programs and slides. - Transparences. - Manual of solved problems (answer and solutions) - Text Books: - Introduction to electrodynamics by David J. Griffiths Prentice Hall Int.	- Lectures. - Problems and essay assignments.	6hr 5hr 5hr 5hr 5hr	1- Maxwell's equations and their solutions. 2- Gauge invariance, Lorentz condition and equation of potential. 3- Microscopic Maxwell's equations: general wave equation, plane wave in nonconductivity medium. 4- Polarization and boundary conditions for emission of electromagnetic radiation. 5- Oscillatory electric dipole and radiation field	The student have to be acquainted with: 1- Maxwell's equations and their solutions. 2- Gauge invariance, and Lorentz condition. 3- Microscopic Maxwell's equations and plane wave in nonconductivity medium. 4- Polarization and emission of electromagnetic radiation. 5- Oscillatory electric dipole and radiation field	

COURSE SYLLABUS محتويات المقرر

الفصل الدراسي الثاني

P421

Radiation Physics

2 h lect./w + 2h lab./w

4th year, 2nd semester

COURSE NO.

COURSE TITLE

NUMBER OF HOURS

YEAR & SEMISTER

COURSE DESCRIPTION

1. رقم المقرر :

2. عنوان المقرر :

3. عدد الساعات :

4. الفرقة والفصل الدراسي :

5. وصف المقرر :

The course introduces the students to the interaction of radiation with matter and biological systems besides the radiation detection and dosemetry.

COURSE GOALS:

6. أهداف المقرر

* The students have to recognize the role of external and internal hazards of radiation sources and radiation protection

7. متطلبات السابقة أو المتوازية : PREREQUESITES AND COREQUISITES

التقويم Assessment	المصادر التعليمية Educational Resources	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	الزمن المتوقع لكل موضوع	الموضوع Topic	مخرجات التعلم Intended Learning Outcomes	م
Semester activities including class - Interactions and Quizzes. - Mid-term exam - Lab performance evaluation. - Final exam.	- Computer simulation programs and slides. - Transparencies. - Manual of solved problems (answer and solutions) - Text books: 1- أسس الفيزياء الإشعاعية، د. محمد فاروق احمد - د. احمد محمد السريع. اصدار النشر العربي والمطابع، جامعة الملك سعود بالسعودية 1997. 2- Atoms, Radiation and Radiation protection, James E. Turner. 1985, Interscience publication, John Wiley & Sons, Inc.	- Lectures. - Laboratory experiments. - Problems and essay assignments. - Visits to radiotherapy institutes and centers of radiation protection and safety departments	6hr	1- Interaction of radiation with matter, the interaction between charged particles, gamma radiation and neutrons with matter.	The student have to recognize the following: 1- Interaction of radiation with matter.	
			4hr	2- Methods of radiation detection and radiation detectors. The gas filled detectors, Scintillation detectors, Solid state detectors, Spark counter, Cherenkof detector, photographic plates.	2- Radiation detection and detectors.	
			4hr	3- Biological effects of ionizing radiations, genetic and somatic effects.	3- Biological effects of radiation.	
			4hr	4- Units of radiation doses, dosimeters and survey meters.	4- Radiation doses and dosimetry.	
			4hr	5- External and internal hazards of radiation	5- External and internal hazards of radiation	

التقويم Assessment	المصادر التعليمية Educational Resources	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	الزمن المتوقع لكل موضوع	الموضوع Topic	مخرجات التعلم Intended Learning Outcomes	م
Semester activities including class - Interactions and Quizzes. - Mid-term exam - Lab performance evaluation. - Final exam.	- Computer simulation programs and slides. - Transparencies. - Manual of solved problems (answer and solutions) - Text Books: - Introduction to electrodynamics by David J. Griffiths Prentice Hall Int.	- Lectures. - Problems and essay assignments.	6hr 5hr 5hr 5hr 5hr	1- Maxwell's equations and their solutions. 2- Gauge invariance, Lorentz condition and equation of potential. 3- Microscopic Maxwell's equations: general wave equation, plane wave in no conducting medium. 4- Polarization and boundary conditions for emission of electromagnetic radiation. 5- Oscillatory electric dipole and radiation field	The student have to be acquainted with: 1- Maxwell's equations and their solutions. 2- Gauge invariance, and Lorentz condition. 3- Microscopic Maxwell's equations and plane wave in nonconductivity medium. 4- Polarization and emission of electromagnetic radiation. 5- Oscillatory electric dipole and radiation field	

التقويم Assessment	المصادر التعليمية Educational Resources	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	الزمن المتوقع لكل موضوع 4hr	الموضوع Topic	مخرجات التعلم Intended Learning Outcomes	م
				sources and radiation protection, treatment of contaminated persons. 6- Radiation contamination, storage of radioactive materials and disposal of radioactive wastes.	sources. 6- Storage of radioactive materials and disposal of radioactive wastes.	

COURSE SYLLABUS المحتويات المقرر

الفصل الدراسي الثاني

P422	<u>COURSE NO.</u>	1. رقم المقرر :
New and renewable energy sources	<u>COURSE TITLE</u>	2. عنوان المقرر :
2 h lect./w + 2h labt./w	<u>NUMBER OF HOURS</u>	3. عدد الساعات :
4 th year, 2 nd semester	<u>YEAR & SEMISTER</u>	4. الفرقة والفصل الدراسي :-
	<u>COURSE DESCRIPTION</u>	5. وصف المقرر :

The course introduces the students to basic renewable energy sources including solar energy, wind energy and that produced from water resources.

6. أهداف المقرر
- COURSE GOALS:
- * Students have to know the essential role of renewable energy as clean safe and some what low cost sources

7. متطلبات السابقة أو المتوازية : PREREQUISITES AND COREQUISITES

التقويم Assessment	المصادر التعليمية Educational Resources	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	الزمن المتوقع لكل موضوع	الموضوع Topic	مخرجات التعلم Intended Learning Outcomes	م
Semester activities including class - Interactions and Quizzes. - Mid-term exam - Lab performance evaluation. - Final exam.	- Computer simulation programs and slides. - Transparences. - Manual of solved problems (answer and solutions) - Text books: 1- "Renewable energy" Editor God Fray boyle; Oxford Univ. Press. 2- "Principle of Environmental physics." By Moneteith, J.L. and Unsworth M.	- Lectures. - Laboratory experiments. - Problems and essay assignments	2hr 2hr 2hr 10hr 4hr 6hr	1- Types of traditional energy sources. 2- Disadvantage of traditional energy sources. 3- Types of renewal energy sources. 4- Solar energy: - Collectors and Concentrators. - Solar selective coating and green house effect. - Solar cells systems. - Storage of solar energy 5- Wind energy generators. 6- Energy produced from water resources: - A Tide and waves energy. - Temperature gradient of water resources. - Water falls	The student have to recognize the following: 1- Traditional vs renewable energy sources. 2- Solar energy. 3- Wind energy generators. 4- Energy produced from water resources.	

COURSE SYLLABUS المقررات

الفصل الدراسي الثاني

P423

Material science

2 h lect./w + 2h lab./w

4th year, 2nd semester

COURSE NO.

فيزياء المواد COURSE TITLE

NUMBER OF HOURS

YEAR & SEMISTER

COURSE DESCRIPTION

1. رقم المقرر :

2. عنوان المقرر :

3. عدد الساعات :

4. الفرقة والفصل الدراسي :

5. وصف المقرر :

The course introduces the students to new and advanced materials and their applications in high technology and daily life.

COURSE GOALS:

6. أهداف المقرر

- * Developing the students creative thought needed to participate in the development of materials science and physics of materials

7. متطلبات السابقة أو المتوازية : PREREQUISITES AND COREQUISITES

التقويم Assessment	المصادر التعليمية Educational Resources	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	الزمن المتوقع لكل موضوع	الموضوع Topic	مخرجات التعلم Intended Learning Outcomes	م
Semester activities including class - Interactions and Quizzes. - Mid-term exam - Lab performance evaluation. - Final exam.	- Computer simulation programs and slides. - Transparencies. - Manual of solved problems (answer and solutions) - Text books: - Material Science and Engineering, an introduction by William D. Callister Jr. John Wiley & Sons.	- Lectures. - Laboratory experiments. - Problems and essay assignments	3hr 2hr 1hr 4hr 2hr 4hr 2hr 2hr 2hr 2hr	1- Structure of metals: close packed arrangement and types of alloys. 2- Equilibrium diagrams. 3- Diffusion of solids. 4- Fundamental of mechanical properties. 5- Theory of plasticity 6- Materials testing: Tension, hardness, fatigue and creep. 7- Deformation of polycrystalline materials. 8- Grain boundaries and deformation. 9- Strengthening mechanisms. 10- Heat treatment: recovery, crystallization and grain growth.	The student have to recognize the following: 1- Structural of metals and alloys. 2- Diffusion in solids. 3- Mechanical properties of solids and their tests. 4- Deformation mechanisms and heat treatment.	

م	مخرجات التعلم Intended Learning Outcomes	الموضوع Topic	الزمن المتوقع لكل موضوع	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	المصادر التعليمية Educational Resources	التقويم Assessment
	5- New and advanced materials.	11- Superconductors, ceramics, liquid-crystals, composite, polymers, nano-materials, photonic materials.	4hr			

محتويات المقرر
COURSE SYLLABUS
الفصل الدراسي الثاني

P424	<u>COURSE NO.</u>	1. رقم المقرر :
Digital and Logic circuits	<u>COURSE TITLE</u>	2. عنوان المقرر :
2 h lect./w + 2h lab./w	<u>NUMBER OF HOURS</u>	3. عدد الساعات :
4 th year, 2 nd semester	<u>YEAR & SEMISTER</u>	4. الفرقة والفصل الدراسي :-
	<u>COURSE DESCRIPTION</u>	5. وصف المقرر :

The course introduces the student to the principles and constructions of logic and digital circuits.

COURSE GOALS:

6 . أهداف المقرر

* The students have to know basic difference between analog and digital electronics and their converters. Developing the students skills in the use of circuits of binary systems.

7. متطلبات السابقة أو المتوازية : PREREQUESITES AND COREQUISITES

التقويم Assessment	المصادر التعليمية Educational Resources	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	الزمن المتوقع لكل موضوع	الموضوع Topic	مخرجات التعلم Intended Learning Outcomes	م
- Semester activities including classroom interactions and quizzes. - Mid-term exam - Lab performance evaluation. - Oral exam. - Final exam.	- Computer simulation programs and slides. - Transparencies. - Manual of solved problems (answer and solutions) - Text books: 1- Microprocessors and microcomputers by D.D.Givon and R.P Roeser, McGraw Hill. 2- " Digital book" by R.M Morston; Newnef.	- Lectures. - Laboratory experiments. - Problems and essay assignments	2hr 4hr 4hr 4hr 4hr 8hr	1- Introduction. 2- Analog electronics- digital electronics – comparison between analog and digital electronics. 3- Convertors Analog to digital (AD) and Digital to analog (DA). 4- Principle of logic circuits. 5- Digital logic circuits: - circuits of binary system. (a) AND - Logic circuit. (b) OR - Logic circuit. (c) NOT - Logic circuit (d) NOR - Logic circuit (e) NAND- Logic circuit	The student have to recognize the following: 1- Analog versus digital electronics and the corresponding converters. 2- Principles of logic circuit. 3- Basic elements of binary systems	

COURSE SYLLABUS المحتويات المقرر

الفصل الدراسي الثاني

P425

Magnetic Resonance

2 h lect./w + 1h tut./w

4th year, 2nd semester

COURSE NO.

الرنين المغناطيسي COURSE TITLE

NUMBER OF HOURS

YEAR & SEMISTER

COURSE DESCRIPTION

This course introduces the students to the basic technical applications of electron spin and magnetic moments (EMR) and nuclear spin and magnetic moment (NMR), some applications of magnetic resonance (MRI) in Biology, and Medicin.

COURSE GOALS:

- * Developing the student skills in the field of nuclear magnetic resonance that provide a good orientation of students needed to meet high technology challenges

PREREQUESITES AND COREQUISITES : متطلبات السابقة أو المتوازية :

1. رقم المقرر :

2. عنوان المقرر :

3. عدد الساعات :

4. الفرقة والفصل الدراسي :-

5. وصف المقرر :

6 . أهداف المقرر

التقويم Assessment	المصادر التعليمية Educational Resources	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	الزمن المتوقع لكل موضوع	الموضوع Topic	مخرجات التعلم Intended Learning Outcomes	م
- Semester activities including classroom interactions and quizzes. - Mid-term exam - Lab performance evaluation. - Final exam.	- Computer simulation programs and slides. - Transparencies. - Manual of solved problems (answer and solutions) - Text books: 1- "Electron spin resonance theory and applications" by N.M.A theraton John Wiley and Sons Inc, New York (1973)". 2- "Electron spin resonance elementary theory and practical applications" by John E.Wentz and James R.Botton McGraw- Hill Book company, New York		2hr 2hr 2hr 4hr	1- The resonance phenomenon. 2- Electron spin and magnetic moment. 3- Energy of free electron in a magnetic field. 4- Electron paramagnetic resonance spectra. 5- Hamiltonian and wave function. 6- The complete Hamiltonian operator. 7- The Zeeman hyperfine. 8- Spin- spin interaction, crystal field. 9- Spin- orbit coupling. 10- Nuclear Zeeman effects.	The student have to recognize the following: 1- Electron paramagnetic resonance spectra. 2- Zeeman hyperfine spin-spin interaction and nuclear Zeeman Effect.	

التقويم Assessment	المصادر التعليمية Educational Resources	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	الزمن المتوقع لكل موضوع	الموضوع Topic	مخرجات التعلم Intended Learning Outcomes	م
	(1971)".		4hr	11- Theory of relation and line shapes.	3- Nuclear magnetic resonance spectrum	
			2hr	12- Electron spin echo. 13- The NMR spectrum.		
			4hr	14- Origin of the spectrum 15- The chemical shift.		
			4hr	16- Physical mechanism underlying chemical shifts line multiplicity in NMR.		
			4hr	17- Analysis of spin systems.		
			4hr	18- The linewidths of NMR spectra.		
			2hr	19- Magnetic noise. 20- The Bloch formulations.	4- Magnetic noise and Bloch formulations.	

COURSE SYLLABUS محتويات المقرر
الفصل الدراسي الثاني

P 426

Research Essay

2 h lect./w + 1h tut./w

4th year, 2nd semester

COURSE NO.

COURSE TITLE مقال وبحث

NUMBER OF HOURS

YEAR & SEMISTER

COURSE DESCRIPTION

1. رقم المقرر :

2. عنوان المقرر :

3. عدد الساعات :

4. الفرقة والفصل الدراسي :-

5. وصف المقرر :

The course introduces the students to make a practice of how to convey out a research work

COURSE GOALS:

6 . أهداف المقرر

* Developing the ability of students for orientation towards research resources.

7. متطلبات السابقة أو المتوازية : PREREQUESITES AND COREQUISITES

م	مخرجات التعلم Intended Learning Outcomes	الموضوع Topic	الزمن المتوقع لكل موضوع	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	المصادر التعليمية Educational Resources	التقويم Assessment
	To make a practice of how the students have reserved a topic and undertook it. Also how the student present and discuss the topics.	Each student work on independent project under the supervision of physics staff member. The essay should be a review paper. Students participate in a series of seminar meeting in which they present a talk on their projects.	26hr	- Seminars. - Discussions. - Library	- M.Sc & Ph.D. thesis. - Library. - Internet. - Text books to be specified by the staff members.	- Discussion in Seminars. - Final report. - Oral Exam

محتويات المقرر COURSE SYLLABUS

الفصل الدراسي الثاني

١. رقم المقرر :	P 427	<u>COURSE NO.</u>
٢. عنوان المقرر :	تاريخ وفلسفة العلم	<u>COURSE TITLE</u>
٣. عدد الساعات :	٢	<u>NUMBER OF HOURS</u>
٤. الفرقة والفصل الدراسي :-	الرابعة - الفصل الدراسي الثاني	<u>YEAR & SEMISTER</u>
٥. وصف المقرر :		<u>COURSE DESCRIPTION</u>

يَهَنَمُ المقرر بتنمية التفكير العلمي عند المتعلم، ويدرك صلة العلم بالسياق الاجتماعي، وأهمية تطبيق العلم في الحياة.

COURSE GOALS:

٦. أهداف المقرر
- تنمية التفكير العلمي لدى الطلاب
 - ربط موضوعات الدراسة ونظريات العلم بسياقها التاريخي الشامل
 - التعرف على فلسفة العلوم
 - الوقوف على أهمية فلسفة العلوم في الحياة.

٧. متطلبات السابقة أو المتوازية : PREREQUISITES AND COREQUISITES

م	مخرجات التعلم Intended Learning Outcomes	الموضوع Topic	الزمن المتوقع لكل موضوع	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	المصادر التعليمية Educational Resources	التقويم Assessment
	- يتبع خطوات التفكير العلمي في حياته ويطبق مناهج البحث العلمي. - يعرف صلة العلم بالمجتمع وكيف يتفاعل العلم مع السياق المجتمعي. - يربط التطور العلمي بالحضارة والتقدم الثقافي. - يهتم بالفكر الفلسفي. - يعرف تطبيقات العلم في حياة البشر.	- نظرية المعرفة وخصائص المعرفة العلمية. - العلاقة المتبادلة بين العلم والتكنولوجيا والمجتمع - مراحل تاريخ العلم ونظرياته. - المنهج العلمي (الرياضي - التجريبي - المعاصر) - التعريف بالإنجازات العلمية والنفسية لعلماء الحضارة العربية الإسلامية في سياقها التاريخي. - تحليل التطور التاريخي للعلوم والنظريات العلمية المعاصرة من خلال نماذج منتقاه من العلوم الأساسية		- بحوث ومقالات يقوم بها الطلاب - زيارات للمتاحف العلمية - تكليفات عن صلة النظريات العلمية بالحياة والمجتمع.	- Theory of knowledge and epistemology of science. - Mutual relation between science, technology and society. - Scientific and technological achievements of Islamic and Arabic Civilization. - Analysis of the historical development of basic sciences (selected examples) - أحمد فؤاد باشا، التراث العلمي للحضارة الإسلامية ومكانته في تاريخ العلم والحضارة - القاهرة	- اختبارات مقال نصف فصلية وفي نهاية الفصل - ملف الطالب

م	مخرجات التعلم Intended Learning Outcomes	الموضوع Topic	الزمن المتوقع لكل موضوع	الأنشطة والواجبات والتعيينات Activities, Tasks and Assignment	المصادر التعليمية Educational Resources	التقويم Assessment
					١٩٨٤ - مقدمة تاريخ العلم - جورج سارتون - قدرى حافظ طوقان، العلوم عند العرب - مناهج البحث العلمى، د. عبد الرحمن بدوى - دار النهضة العربية ١٩٩٨. - دونالد هيل، العلوم والهندسة فى الحضارة الاسلامية، ترجمة أحمد فؤاد باشا - سلسلة عالم المعرفة - الكويت ٢٠٠٤م	